



## Practice Assignment 6 - Not Graded



This assignment will not be graded and is only for practice.

### Multiple Choice Questions (MCQ):

1 point

Which of the following statements are correct?

☐

There exists a linear transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ ,  $T(1, -1) = (1, -1)$ , and  $T(4, 1) = (1, 0)$ .

☐

If there is a linear transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ , and  $T(1, -1) = (1, -1)$ , then  $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$ .

☐

Let  $\beta = \{v_1, v_2, v_3\}$  and  $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be bases of a vector space  $V$ . Consider a linear transformation  $T: V \rightarrow V$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . The matrix representation of  $T$ , with respect to  $\beta$  and  $\gamma$  for the domain and codomain respectively, is

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 2 \\ 1 & -2 & 3 \end{bmatrix}$$

☐

Let  $\{v_1, v_2, v_3\}$  be a basis of a vector space  $V$  and  $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be a basis of vector space  $W$ . Consider a linear transformation  $T: V \rightarrow W$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . Then, the rank of  $T$  is 3.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

If there is a linear transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $T(2, 3) = (1, 2)$ , and  $T(1, -1) = (1, -1)$ , then  $T(x, y) = \frac{1}{5}(4x - y, -x + 4y)$ .

Let  $\beta = \{v_1, v_2, v_3\}$  and  $\gamma = \{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be bases of a vector space  $V$ . Consider a linear transformation  $T: V \rightarrow V$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . The matrix representation of  $T$ , with respect to  $\beta$  and  $\gamma$  for the domain and codomain respectively, is

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 0 & 2 \\ 1 & -2 & 3 \end{bmatrix}$$

Let  $\{v_1, v_2, v_3\}$  be a basis of a vector space  $V$  and  $\{2v_1 + v_3, v_2 - v_3, v_3 - v_1\}$  be a basis of vector space  $W$ . Consider a linear transformation  $T: V \rightarrow W$  such that  $T(v_1) = 2v_3 + v_1$ ,  $T(v_2) = 2v_1$ , and  $T(v_3) = 2v_2$ . Then, the rank of  $T$  is 3.



1 point

Consider the following set  $S = \{(x, y) \mid x + y = 1, x, y \in \mathbb{R}\}$ . In column A the matrix representation of some linear transformations are given with respect to the standard ordered bases. Match the entries in column A with the set  $T(S)$  given in column B and the geometric representations of  $SS$  and  $T(S)$  in column C.

☐

i  $\rightarrow$   $\rightarrow$  b  $\rightarrow$   $\rightarrow$  2.

☐

ii  $\rightarrow$   $\rightarrow$  c  $\rightarrow$   $\rightarrow$  1.

☐

ii  $\rightarrow$   $\rightarrow$  c  $\rightarrow$   $\rightarrow$  2.

☐

iii  $\rightarrow$   $\rightarrow$  a  $\rightarrow$   $\rightarrow$  3.

☐

i  $\rightarrow$   $\rightarrow$  b  $\rightarrow$   $\rightarrow$  3.

☐

iii  $\rightarrow \rightarrow a \rightarrow \rightarrow 1$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

ii  $\rightarrow \rightarrow c \rightarrow \rightarrow 2$ .

i  $\rightarrow \rightarrow b \rightarrow \rightarrow 3$ .

iii  $\rightarrow \rightarrow a \rightarrow \rightarrow 1$ .

### Multiple Select Questions (MSQ):

1 point

Consider the following two groups. In Group 1, there are 3 functions from one vector space to another, and in Group 2, there are three properties of functions. All the vector spaces have usual addition and scalar multiplication.

#### Group-1

- **P:**  $T: R^2 \rightarrow R^2$   $T: R^2 \rightarrow R^2$  such that  $T(x, y) = \begin{cases} (\frac{x^2}{y}, y) & \text{if } y \neq 0 \\ (x, 0) & \text{otherwise} \end{cases}$   $T(x, y) = \{ \mid \} \mid \{ (yx^2, y)$   
 $(x, 0)$  if  $y = 0$  otherwise
- **Q:**  $T: R^2 \rightarrow R^2$   $T: R^2 \rightarrow R^2$  such that  $T(x, y) = (x, xy)$   $T(x, y) = (x, xy)$
- **R:**  $T: R^3 \rightarrow R^3$   $T: R^3 \rightarrow R^3$  such that  $T(x, y, z) = \frac{x+y+z}{5}T(x, y, z) = 5x+y+z$

#### Group-2

- 1:  $T(v_1 + v_2) = T(v_1) + T(v_2)$   $T(v_1 + v_2) = T(v_1) + T(v_2)$  for all  $v_1, v_2 \in V$   $v_1, v_2 \in V$
- 2:  $T(c v) = c T(v)$   $T(c v) = c T(v)$  for all  $v \in V$   $v \in V$  and  $c \in R$   $c \in R$ .
- 3:  $T$  is a linear transformation

Choose the set of correct options.

- ☐ P satisfies 1 but not 2.
- ☐ P satisfies 2 but not 1
- ☐ Q satisfies 1
- ☐ Q satisfies 3
- ☐ P satisfies 3
- ☐ R satisfies 2
- ☐ R satisfies 3
- ☐ R satisfies 1 but not 2

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

P satisfies 2 but not 1

R satisfies 2

R satisfies 3

### Numerical Answer Type (NAT):

Suppose the matrix representation of a linear transformation  $T: R^3 \rightarrow R^3$   $T: R^3 \rightarrow R^3$  with respect to the ordered bases  $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$   $\beta = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$  for the domain and  $\gamma = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$   $\gamma = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$  for the range, is  $I_{3 \times 3}$   $I_{3 \times 3}$ , i.e., the

identity matrix of order 3. Let  $AA$  be the matrix representation of the linear transformation  $TT$  with respect to standard ordered basis of  $R^3 R3$  for both domain and range. Then the determinant of  $AA$  is

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

Let  $TT$  and  $SS$  be two linear transformations from  $R^3 R3$  to  $R^3 R3$ , which are defined as follows:

$$T: R^3 \rightarrow R^3 \quad T: R3 \rightarrow R3 \quad S: R^3 \rightarrow R^3 \quad S: R3 \rightarrow R3$$

$$T(x, y, z) = (x - y, y - z, z - x) \quad T(x, y, z) = (x - y, y - z, z - x)$$

$$S(x, y, z) = (x + y, y + z, z + x) \quad S(x, y, z) = (x + y, y + z, z + x)$$

Let  $S + TS + T$  be defined to be the linear transformation as follows:

$$(S + T): R^3 \rightarrow R^3 \quad (S + T): R3 \rightarrow R3$$

$$(S + T)(x, y, z) = S(x, y, z) + T(x, y, z) \quad (S + T)(x, y, z) = S(x, y, z) + T(x, y, z)$$

Let  $CC$  be the matrix representation of  $S + TS + T$  with respect to the standard ordered bases of  $R^3 R3$ . If  $C = nI$ , where  $II$  denotes the identity matrix of order 3, then what will be the value of  $n$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

1 point

Consider a linear transformation  $T: R^3 \rightarrow R^3 \quad T: R3 \rightarrow R3$  defined as

$T(x, y, z) = (x + 5y, z - 3x, y + 6z) \quad T(x, y, z) = (x + 5y, z - 3x, y + 6z)$ . Which of the following options are true about  $TT$ ?

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Nullity of  $TT$  is 00.

☐

A basis for the range of  $TT$  is  $\{(1, 5, 0), (0, 1, 6)\} \{(1, 5, 0), (0, 1, 6)\}$ .

☐

A basis for the range of  $TT$  is  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ .

☐

Rank of  $TT$  is 22.

☐

The matrix representation of  $TT$  with respect to the standard ordered bases of  $R^3 R3$  is  $\begin{bmatrix} 1 & -30 & 150 \\ 5 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}$

$\begin{bmatrix} -30 & 10 & 16 \end{bmatrix}$ .



The matrix representation of  $T^T$  with respect to the standard ordered bases of  $R^3$  is  $\begin{bmatrix} 1 & 50 \\ -3 & 0 \\ 0 & 16 \end{bmatrix}$ .  
 $\begin{bmatrix} 1 & 50 \\ -3 & 0 \\ 0 & 16 \end{bmatrix}$ .



Nullity of  $T^T$  is 11.



Rank of  $T^T$  is 33.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Nullity of  $T^T$  is 00.

A basis for the range of  $T^T$  is  $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ .

The matrix representation of  $T^T$  with respect to the standard ordered bases of  $R^3$  is  $\begin{bmatrix} 1 & 50 \\ -3 & 0 \\ 0 & 16 \end{bmatrix}$ .

$\begin{bmatrix} 1 & 50 \\ -3 & 0 \\ 0 & 16 \end{bmatrix}$ .

Rank of  $T^T$  is 33.

Let  $V$  be a vector space with ordered basis  $\alpha = \{v_1, v_2, \dots, v_{10}\}$ . Consider a linear transformation  $T: V \rightarrow V$  such that  $T(v_1) = v_1$  and

$T(v_i) = v_i - v_{i-1}$ , where  $i > 1$ . Let  $A$  be the matrix representation of the linear transformation  $T^T$ , with respect to  $\alpha$  for both the domain and codomain. what is the trace of  $A$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 10

1 point

### Comprehension Type Question:

For each video on YouTube, consider the vector (Number of comments in thousands, Number of shares in thousands, Number of views in thousands) and consider the subset  $S$  of  $R^3$  consisting of these vectors. It is known that the ranking of a YouTube video can be thought of as the restriction of a linear transformation  $T: R^3 \rightarrow R$  to  $S$  where  $R^3$  comes with usual addition and scalar multiplication. (assume ranking 0 (zero) is possible and in practice negative rankings can be ignored). The following data in Table M2W6P2 shows the correlation between comments, shares, and views - with YouTube rankings of a video.

Use the above information to answer questions 8, 9 and 10.

1 point

Which of the following sets of vectors in  $SS$  have the property that the YouTube video corresponding to every linear combination of that set which is in  $SS$  has ranking 00?

☐

$\{(1, 1, 0), (0, 0, 1)\} \{(1, 1, 0), (0, 0, 1)\}$

☐

$\{(6, 25, 0), (0, 0, 25)\} \{(6, 25, 0), (0, 0, 25)\}$

☐

$\{(25, 6, 0), (0, 0, 25)\} \{(25, 6, 0), (0, 0, 25)\}$

☐

$\{(12, 50, 0), (0, 0, 1)\} \{(12, 50, 0), (0, 0, 1)\}$

☐

$\{(24, 100, 0), (0, 0, 25)\} \{(24, 100, 0), (0, 0, 25)\}$

☐

$\{(6, 0, 0), (0, 25, 0), (0, 0, 1)\} \{(6, 0, 0), (0, 25, 0), (0, 0, 1)\}$

☐

$\{(25, 0, 0), (0, 6, 0), (0, 0, 1)\} \{(25, 0, 0), (0, 6, 0), (0, 0, 1)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(6, 25, 0), (0, 0, 25)\} \{(6, 25, 0), (0, 0, 25)\}$

$\{(12, 50, 0), (0, 0, 1)\} \{(12, 50, 0), (0, 0, 1)\}$

$\{(24, 100, 0), (0, 0, 25)\} \{(24, 100, 0), (0, 0, 25)\}$

1 point

Which of the following sets of vectors in  $SS$  have the property that the YouTube video corresponding to every linear combination of that set which is in  $SS$  has ranking a multiple of 1515?

☐

$\{(3, 0, 0), (0, 0, 1)\} \{(3, 0, 0), (0, 0, 1)\}$

☐

$\{(3, 0, 1)\} \{(3, 0, 1)\}$

☐

$\{(3, 0, 1), (0, 0, 1)\} \{(3, 0, 1), (0, 0, 1)\}$

☐

$\{(0, 0, 1)\} \{(0, 0, 1)\}$

☐

$\{(9, 25, 0), (0, 0, 1)\} \{(9, 25, 0), (0, 0, 1)\}$

☐

$\{(25, 9, 0), (0, 0, 1)\} \{(25, 9, 0), (0, 0, 1)\}$

☐

$\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\} \{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$

☐

$\{(25, 0, 0), (0, 9, 0), (0, 0, 1)\} \{(25, 0, 0), (0, 9, 0), (0, 0, 1)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(3, 0, 0), (0, 0, 1)\} \{(3, 0, 0), (0, 0, 1)\}$

$\{(3, 0, 1)\} \{(3, 0, 1)\}$

$\{(3, 0, 1), (0, 0, 1)\} \{(3, 0, 1), (0, 0, 1)\}$ 
 $\{(0, 0, 1)\} \{(0, 0, 1)\}$ 
 $\{(9, 25, 0), (0, 0, 1)\} \{(9, 25, 0), (0, 0, 1)\}$ 
 $\{(9, 0, 0), (0, 25, 0), (0, 0, 1)\} \{(9, 0, 0), (0, 25, 0), (0, 0, 1)\}$ 

Suppose that for a YouTube video there are 5 (in thousands) shares. What must be the number of comments (in thousands) so that the ranking of the video will be 4? [Note: Suppose your answer is 6000, then you have to enter 6 as the answer.]

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

*1 point*

[Check Answers](#)